Resoluções Lista 2 - Álgebra Linear 01.08

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2. $T: V \to W$ $T(u) = 3u$ $T(v) = u - v$

a) $T(u+v)$

$T(u+v) = T(u) + T(v)$

$T(u+v) = 3u + (u - v)$

$T(u+v) = \underline{4u - v}$

b) $T(3v)$

$T(\alpha v) = \alpha T(v)$

$T(3v) = 3T(v)$

$T(3v) = 3.(u-v)$

$T(3v) = \underline{3u - 3v}$

c) $T(4u - 5v)$

$T(\alpha u - \alpha v) = \alpha T(u) - \alpha T(v)$

$4T(u) - 5T(v)$

$4.3u - 5(u-v)$

$12u - 5u + 5v$

$\underline{7u + 5v}$

$\boxed{\begin{array}{l} R: 4u - v \\ R = 3u - 3v \\ R = 7u + 5v \end{array}}$

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3. $T: \mathbb{R}^2 \to \mathbb{R}^2$

a) $T(x,y) = (x - 3y, 2x + 5y)$

$T(x + y) = T(x) + T(y)$

$T(x_1 + x_2, y_1 + y_2) = ((x_1 + x_2) + 2(x_1 + x_2), -3(y_1 + y_2) + 5(y_1 + y_2))$

$= (x_1 + x_2 + 2x_1 + 2x_2, -3y_1 - 3y_2 + 5y_1 + 5y_2)$

$T(x) = T(x_1, y_1) = (x_1 - 3y_1, 2x_1 + 5y_1)$

$T(y) = T(x_2, y_2) = (x_2 - 3y_2, 2x_2 + 5y_2)$

$T(u) + T(v) = (x_1 - 3y_1 + x_2 - 3y_2, 2x_1 + 5y_1 + 2x_2 + 5y_2)$

$T(u) + T(v) = (x_1 + x_2 - 3y_1 - 3y_2, 2x_1 + 2x_2 + 5y_1 + 5y_2)$

$T(\alpha u) = cT(u)$

$T(\alpha x) = \alpha T(x_1, y_1)$

$T(x_1, y_1) = \alpha(x_1 - 3y_1, 2x_2 + 5y_1)$

$T(\alpha x) = T(\alpha x_1, \alpha y_1)$

$= (\alpha x_1 - \alpha 3y_1, \alpha 2x_2 + \alpha 5y_1)$

$= \alpha(x_1 - 3y_1, 2x_2 + 5y_1)$

$\boxed{\text{é linear}}$

b) $T(x,y) = (y,x)$

$T(x+y) = T(x) + T(y)$

$T(x_1 + x_2, y_1 + y_2) = (y_1 + y_2, x_1 + x_2)$

$T(x) = (y_1, x_1)$

$T(y) = (y_2, x_2)$

$T(x+y) = (y_1 + y_2, x_1 + x_2)$

$T(\alpha u) = \alpha T(u)$

$T(\alpha x_1, \alpha y_1) = (\alpha x_1, \alpha y_1)$

$\alpha T(u) = \alpha(x_1, y_1)$

$\boxed{\text{é linear}}$

c) $T(x,y) = (x^2, y^2)$

$u = (x_1, y_1)$

$v = (x_2, y_2)$

$T(u+v) = T(u) + T(v)$

$T(x_1 + x_2, y_1 + y_2) = ((x_1 + x_2)^2, (y_1 + y_2)^2)$

$T(u) = T(x_1, y_1) = (x_1^2, y_1^2)$

$T(v) = T(x_2, y_2) = (x_2^2, y_2^2)$

$T(u+v) \neq T(u) + T(v)$

$(x_1 + x_2)^2 \neq x_1^2 + x_2^2$ //

$T(\alpha u) = \alpha T(u)$

$T(\alpha x_1, \alpha y_1) = \underline{\alpha T(x_1, y_1)}$

$T(\alpha x_1, \alpha y_1) \neq \alpha(x_1^2, y_1^2)$

$\boxed{\text{não é linear}}$

d) $T(x,y) = (x+1, y)$

$T(u + v) = T(u) + T(v)$

$u = (x_1, y_1)$

$v = (x_2, y_2)$

$T(x_1 + y_1, x_2 + y_2) = T(x_1 + y_1) + T(x_2 + y_2)$

$T(x_1 + 1 + y_1, x_2 + 1 + y_2) = T(x_1 + 1 + y_1) + T(x_2 + 1, y_2)$

$\neq (x_1 + x_2 + 1, y_1 + y_2)$

$T(c(x,y)) = T(cx, cy)$

$c(x_1 + 1, y_1) \neq T(cx_1 + 1, cy_1)$

$\boxed{\text{não é linear}}$

$\boxed{\begin{array}{l} \text{Gabarito:} \\ \text{a) Linear} \\ \text{b) Linear} \\ \text{c) Não Linear} \\ \text{d) Não Linear} \end{array}}$

e) $T(x,y) = (y-x, 0)$

$T((x_1,y_1)+(x_2,y_2)) = T(x_1+x_2, y_1+y_2)$

$= (y_1+y_2-x_1-x_2, 0)$

$= (y_1-x_1, 0) + (y_2-x_2, 0)$

$T(x_1,y_1) + T(x_2,y_2)$

$T(c(x,y)) = T(cx, cy)$

$= (cy-cx, 0)$

$c(y-x, 0)$

$cT(x,y)$

$\boxed{\text{é linear}}$

f) $T(x,y) = (|x|, 2y)$ $\quad T(u+v) = T(u) + T(v)$

$T((x_1,y_1)+(x_2,y_2)) = T(x_1+x_2, y_1+y_2)$

$= (|x_1+x_2|, 2(y_1+y_2)) \neq (|x_1|, 2y_1) + (|x_2|, 2y_2)$

$T(\alpha u) = \alpha T(u)$ $\quad T(u) = T(|x_1|, 2y_1)$ $\quad T(\alpha u) \neq \alpha T(u)$

$u = (|x_1|, 2y_1)$

$T(|\alpha x_1|, 2\alpha y_1)$

$\alpha T(u) = \alpha(|x_1|, 2y_1)$

$\boxed{\text{não é linear}}$

g) $T(x,y) = (\operatorname{sen} x, y)$

$u = (x_1, y_1)$

$v = (x_2, y_2)$

$T(u+v) = T(u) + T(v)$

$T((x_1+y_1)+(x_2+y_2)) = T(x_1+y_1) + T(x_2,y_2)$

$T((\operatorname{sen} x_1 + y_1) + (\operatorname{sen} x_2, y_2)) = T(\operatorname{sen} x_1, y_1) + T(\operatorname{sen} x_2, y_2)$

$T(\alpha u) = \alpha T(u)$

$T(\alpha(x,y)) = T(\alpha x, \alpha y)$

$\operatorname{sen} \alpha x, \alpha y = \alpha(\operatorname{sen} x, y)$

$\underline{\operatorname{Sen} \alpha x \neq \alpha \operatorname{sen} x}$

$\boxed{\text{não é linear}}$

Gabarito

e) Linear

f) Não é Linear

g) Não é linear

h) Não é linear

i) Linear

h) $T(x,y) = (xy, x-y)$

$T(u+v) = T(u) + T(v)$

$T((x_1,y_1)+(x_2,y_2)) = T(x_1+x_2, y_1+y_2)$

$= ((x_1+x_2)(y_1+y_2)), (x_1+x_2)-(y_1-y_2))$

$T(u) = (x_1y_1, x_1-y_1)$

$T(v) = (x_2y_2, x_2-y_2)$ $\; + \; (x_1y_1+x_2y_2, (x_1-y_1)+(x_2-y_2))$

$\boxed{\text{não é linear}}$

$T(\alpha u) = \alpha T(u)$

$(\alpha x \alpha y, \alpha x - \alpha y) = \alpha T(x_1y_1, x_1-y_1)$

$(\alpha x \alpha y, \alpha x - \alpha y) \neq (\alpha xy, \alpha(x-y))$

i) $T(x,y) = (3y, -2x)$

$T(u) = T(x_1, y_1)$

$T(v) = T(x_2, y_2)$ $\; + \; = (3y_1, -2x_1) + (3y_2, -2x_2)$

$(3y_1+3y_2, -2x_1-2x_2)$

$T(u+v) = T(x_1+x_2, y_1+y_2)$

$T(u+v) = (3(y_1+y_2), -2(x_1+x_2))$

$(3(y_1+y_2), -2(x_1+x_2)) = (3y_1+3y_2, -2x_1-2x_2)$

$\boxed{\text{É Linear}}$

$T(\alpha u) = \alpha T(u)$

$T(\alpha x, \alpha y) = \alpha(x,y)$

$(3\alpha y - 2\alpha x) = (3\alpha y - 2\alpha x)$

5. a) $T: \mathbb{R}^2 \to \mathbb{R}^3$ $\quad T(x,y) = (x-y, 3x, -2y)$

$T(u+v) = T(x_1+x_2, y_1+y_2)$

$= (x_1+x_2-(y_1+y_2), 3(x_1+x_2), -2(y_1+y_2))$

$= (x_1-y_1+x_2-y_2, 3x_1+3x_2, -2y_1-2y_2)$

$T(u) = (x_1-y_1, 3x_1, -2y_1)$

$T(v) = (x_2-y_2, 3x_2, -2y_2)$ $\; + \; (x_1-y_1+x_2-y_2, 3x_1+3x_2, -2y_1-2y_2)$

$\boxed{\text{é linear}}$

$T(\alpha u)$

$= (\alpha x - \alpha y, 3\alpha x, -2\alpha y)$

$\alpha T(u) = \alpha(x-y), \alpha(3x), \alpha(-2y)$

$= (\alpha x - \alpha y, 3\alpha x, -2\alpha y)$

$\underline{T(\alpha u) = \alpha T(u)}$

b) $T: \mathbb{R}^3 \to \mathbb{R}^3$ $\quad T(x,y,z) = (x+y, x-y, 0)$

$T(u+v) = ((x_1+x_2)+(y_1+y_2), (x_1+x_2)-(y_1+y_2), 0)$

$T(u) = (x_1+y_1, x_1-y_1, 0)$

$T(v) = (x_2+y_2, x_2-y_2, 0)$ $\; + \; (x_1+y_1+x_2+y_2, x_1-y_1+x_2-y_2, 0)$

$\boxed{\text{é linear}}$

$T(u+v) = T(u) + T(v)$

$T(\alpha u) = (\alpha x + \alpha y, \alpha x - \alpha y, 0)$

$\alpha T(u) = \alpha(x+y), \alpha(x-y), 0)$

$\underline{T(\alpha u) = \alpha T(u)}$

c) $T: \mathbb{R}^2 \to \mathbb{R}^2 \quad T(x,y) = (x^2+y^2, x)$

$T(u+v) = ((x_1+x_2)^2 + (y_1+y_2)^2, x_1+x_2)$

$\left.\begin{array}{l} T(u) = (x_1^2+y_1^2, x_1) \\ T(v) = (x_2^2+y_2^2, x_2) \end{array}\right\} + = (x_1^2+x_2^2+y_1^2+y_2^2, x_1+x_2)$

$\neq$ Não é linear

d) $T: \mathbb{R} \to \mathbb{R}^2 \quad T(x) = (x, 2)$

$u = x_1$

$v = x_2$

$T(u+v) = T(x_1+x_2) = (x_1+x_2, 2) \neq$

$\left.\begin{array}{l} T(u) = (x_1, 2) \\ T(v) = (x_2, 2) \end{array}\right\} + = (x_1+x_2, 4)$

Não é linear

$T(\alpha u) = (\alpha x, 2)$

$\alpha T(u) = (\alpha x, \alpha 2)$

$\neq$

e) $T: \mathbb{R}^3 \to \mathbb{R} \quad T(x,y,z) = -3x+2y-z$

$T(u+v) = -3(x_1+x_2) + 2(y_1+y_2) - (z_1+z_2)$

$T(u+v) = (-3x_1+2y_1-z_1) + (-3x_2, 2y_2, z_2)$

$T(u) = (-3x_1+2y_1-z_1)$ =

$T(v) = (-3x_2+2y_2-z_2)$

$T(\alpha u) = -3(\alpha x) + 2(\alpha x) - (\alpha z)$

$\alpha T(u) = \alpha(-3x+2y-z)$

=

$\boxed{\text{é Linear}}$

f) $T: \mathbb{R}^2 \to \mathbb{R}^2 \quad T(x,y) = (|x|, y)$

$T(u+v) = (|x_1+x_2|, y_1+y_2)$

$T(u) = (|x_1|, y_1) \neq$

$T(v) = |(x_2|, y_2)$

$\boxed{\text{Não é linear}}$

$\alpha T(u) = (\alpha|x|, \alpha y)$

$T(\alpha u) = (|\alpha x|, \alpha y)$

$\neq$

g) $T: \mathbb{R}^2 \to \mathbb{R} \quad T(x,y) = x$

$T(u+v) = x_1+x_2$

$\left.\begin{array}{l} T(u) = x_1 \\ T(v) = x_2 \end{array}\right\} = T(u+v)$

$\boxed{\text{é Linear}}$

$\alpha T(u) = \alpha x$

$T(\alpha u) = \alpha x$

=

h) $T: \mathbb{R}^2 \to \mathbb{R} \quad T(x,y) = xy$

$T(u+v) = (x_1+x_2)(y_1+y_2)$

$T(u) = (x_1y_1)$ + = $x_1y_1 + x_2y_2$

$T(v) = (x_2y_2)$ $\neq$

$(x_1+x_2)(y_1+y_2)$

$\boxed{\text{não é linear}}$

$T(\alpha u) = (\alpha x)(\alpha y)$

$\alpha T(u) = \alpha xy$

$\neq$

i) $T: \mathbb{R}^2 \to \mathbb{R}^4 \quad T(x,y) = (y, x, y, x)$

$T(u+v) = (y_1+y_2, x_1+x_2, y_1+y_2, x_1+x_2)$

$T(u) = (y_1, x_1, y_1, x_1)$

$T(v) = (y_2, x_2, y_2, x_2)$ $= T(u+v)$

$\alpha T(u) = (\alpha y, \alpha x, \alpha y, \alpha x)$

$T(\alpha u) = (\alpha y, \alpha x, \alpha y, \alpha x)$

$=$

$\boxed{\text{é linear}}$

Gabarito
i) Linear
j) Linear
k) Linear
l) Não é Linear
m) Linear

j) $T: \mathbb{R}^2 \to M(2,2),\ T(x,y) = \begin{pmatrix} 2y & 3x \\ -y & x+2y \end{pmatrix}$

$T(u+v) = \begin{pmatrix} 2(y_1+y_2) & 3(x_1+x_2) \\ -(y_1+y_2) & (x_1+x_2)+2(y_1+y_2) \end{pmatrix}$

$T(u) = \begin{pmatrix} 2y_1 & 3x_1 \\ -y_1 & x_1+2y_1 \end{pmatrix} \quad T(v) = \begin{pmatrix} 2y_2 & 3x_2 \\ -y_2 & x_2+2y_2 \end{pmatrix}$

$+ = \begin{pmatrix} 2(y_1+y_2) & 3(x_1+x_2) \\ -(y_1+y_2) & (x_1+x_2)+2(y_1+y_2) \end{pmatrix} = T(u+v)$

$\alpha T(u) = \begin{pmatrix} 2\alpha y & 3\alpha x \\ -\alpha y & \alpha(x+2y) \end{pmatrix}$

$T(\alpha u) = \begin{pmatrix} 2\alpha y & 3\alpha x \\ -\alpha y & \alpha(x+2y) \end{pmatrix}$

$\boxed{\text{é linear}}$

k) $T: M(2,2) \to \mathbb{R}^2,\ T\begin{pmatrix} a & b \\ c & d \end{pmatrix} = (a-c, b+c)$

$A = \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} \quad B = \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix}$

$T(A) = (a_1-c_1, b_1-c_1)$

$T(B) = (a_2-c_2, b_2-c_2)$

$T(A)+T(B) = T(A+B)$

$T(A+B) = T\begin{pmatrix} a_1+a_2 & b_1+b_2 \\ c_1+c_2 & d_1+d_2 \end{pmatrix}$

$= (a_1+a_2-c_1-c_2, b_1+b_2+c_2+c_2)$

$T(\alpha A) = T\begin{pmatrix} \alpha a & \alpha b \\ \alpha c & \alpha d \end{pmatrix} = (\alpha a - \alpha c, \alpha b + \alpha c)$

$\alpha T(A) = \alpha(a-c, b+c)$

$(\alpha a - \alpha c, \alpha b + \alpha c)$

$\boxed{\text{é linear}}$

l) $T: M(2,2) \to \mathbb{R} \quad T\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \det \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

$T(A+B) = \det(A+B)$

$T(A)+T(B) = \det(A) + \det(B)$

$T(A+B) \neq T(A)+T(B)$

$\boxed{\text{Não é linear}}$

$T(\alpha A) = \det(\alpha A)$

$\det\begin{pmatrix} \alpha a & \alpha b \\ \alpha c & \alpha d \end{pmatrix} = \alpha^2 \det(A)$

$\alpha T(A) = \alpha \det(A)$

$T(\alpha A) \neq \alpha T(A)$

m) $T: \mathbb{R}^3 \to \mathbb{R}^2 \quad (x,y,z) \to \begin{bmatrix} 2 & 1 & 3 \\ -1 & 0 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

$T(u) = \begin{pmatrix} 2x_1+y_1+3z_1+2x_2+y_2+3z_2 \\ -x_1-2z_1-x_2-2z_2 \end{pmatrix} = \begin{matrix} 2(x_1+x_2)+(y_1+y_2)+3(z_1+z_2) \\ -(x_1+x_2) + 0 - 2(z_1+z_2) \end{matrix}$

$u = (x_1, y_1, z_1)$

$v = (x_2, y_2, z_2)$

$T(u+v) = T(x_1+x_2, y_1+y_2, z_1+z_2)$

$\begin{pmatrix} 2 & 1 & 3 \\ -1 & 0 & -2 \end{pmatrix} \begin{pmatrix} x_1+x_2 \\ y_1+y_2 \\ z_1+z_2 \end{pmatrix} \quad T(u) = \begin{pmatrix} 2x_1+y_1+3z_1 \\ -x_1+0-2z_1 \end{pmatrix} + T(v) = \begin{pmatrix} 2x_2+y_2+3z_2 \\ -x_2+0-2z_2 \end{pmatrix} = T(u+v)$

$= \begin{matrix} 2(x_1+x_2)+(y_1+y_2)+3(z_1+z_2) \\ -(x_1+x_2) + 0 - 2(z_1+z_2) \end{matrix} = \begin{pmatrix} 2x_1+y_1+3z_1 \\ -x_1+0-2z_1 \end{pmatrix} + \begin{pmatrix} 2x_2+y_2+3z_2 \\ -x_2+0-2z_2 \end{pmatrix}$

$T(\alpha u) = T(\alpha x, \alpha y, \alpha z)$

$\begin{pmatrix} 2 & 1 & 3 \\ -1 & 0 & -2 \end{pmatrix} \begin{pmatrix} \alpha x \\ \alpha y \\ \alpha z \end{pmatrix} = \begin{pmatrix} 2\alpha x + \alpha y + 3\alpha z \\ -\alpha x + 0 - 2\alpha z \end{pmatrix}$

$\alpha T(u) = \alpha \begin{pmatrix} 2x+y+3z \\ -x+0-2z \end{pmatrix} = \begin{pmatrix} 2\alpha x + \alpha y + 3\alpha z \\ -\alpha x + 0 - 2\alpha z \end{pmatrix}$

$\boxed{\text{é linear}}$

$$\overset{}{6.\ T:\mathbb{R}^2\to\mathbb{R}^3 \quad (x,y) \to (\overset{x}{x+ky},\ \overset{y}{x+k},\ \overset{z}{y})}$$

a) $k = x$

$(x,y) = (x+xy,\ x+x,\ y)$

$(x,y) = (x(1+y),\ 2x,\ y)$

$T(x_1+x_2,\ y_1+y_2) = ((x_1+x_2)$

$T(x_1,y_1) = (x_1(1+y_1),\ 2x_1,\ y_1)$

$T(x_2,y_2) = (x_2(1+y_2),\ 2x_2,\ y_2)$

$T(x_1,y_1) + T(x_2,y_2) = (x_1(1+y_1) + x_2(1+y),\ 2x_1+2x_2,\ y_1+y_2)$

$= (x_1+x_2)(1+y_1+y_2) = x_1 + x_1y_1 + x_1y_2 + x_2 + x_2y_1 + x_2y_2$

$x_1(1+y_1) + x_2(1+y_2) = x_1 + x_1y_1 + x_2 + x_2y_2$

$\boxed{\text{não é linear}}$

b) $k = 1$

$(x,y) \to (x+1y,\ x+1,\ y)$

i) $T(0,0) = (0+0,\ 0+1,\ 0) = (0,1,0)$

$\boxed{\text{não é transformação linear.}}$

c) $k = 0$

$T(0,0) = (0+0.0,\ 0+0,\ 0) = (0,0,0)$

$T(\vec{u}+\vec{v}) = T(\vec{u}) + T(\vec{v})$

$T(\vec{u}+\vec{v}) = T(x_1+x_2,\ y_1+y_2)$

$T(\vec{u}+\vec{v}) = (x_1+x_2,\ x_1+x_2,\ y_1+y_2)$

$\underbrace{(x_1, x_1, y_1)}_{T(\vec{u})} + \underbrace{(x_2, x_2, y_2)}_{T(\vec{v})}$

$T(\alpha\vec{u}) = \alpha T(\vec{u});\ \alpha \in \mathbb{R}\quad \vec{u}=(x,y)$

$T(\alpha\vec{u}) = T(\alpha x, \alpha y) = ((\alpha x), (\alpha x), (\alpha y))$

$\alpha(x,x,y) \to \boxed{\text{é transformação}}$

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7. a) $T:\mathbb{R}^2\to\mathbb{R}^3\quad T(-1,1) = (3,2,1)$ e $T(0,1) = (1,1,0)$.

$(x,y) = a_1.v_1 + a_2.v_2$

$(x,y) = a_1(-1,1) + a_2(0,1)$

$(x,y) = (-a_1,\ a_1) + a_2$

$-a_1 = x$

$a_1 + a_2 = y \qquad (-x)\vec{v_1} + (y+x)\vec{v_2}$

$a_2 = y - a_1 \qquad T(x,y) = (-x)\,T(\vec{v_1}) + (y+x)\,T(\vec{v_2})$

$a_2 = y+x \qquad T(x,y) = (-x).(3,2,1) + (y+x).(1,1,0)$

$a_1 = -x \qquad T(x,y) = (-3x,\ -2x,\ -x) + (y+x,\ y+x,\ 0) \Rightarrow \boxed{(y-2x,\ y-x,\ -x)}$

b) $v \in \mathbb{R}\quad T(v) = (-2,1,-3)$

$T(v) = (-2,1,-3)$

$T(x,y) = (y-2x,\ y-x,\ -x)$

$$\begin{cases} y-2x = -2 \\ y-x = 1 \\ -x = -3 \end{cases} \quad x = 3 \qquad \begin{aligned} y &= 2x-2 \\ y &= 2.3-2 \\ y &= 4 \end{aligned} \qquad \boxed{v = (\overset{x}{3}, \overset{y}{4})}$$

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8 a) $T:\mathbb{R}^3\to\mathbb{R}^2\quad T(1,-1,0) = (1,1)\quad T(0,1,1) = (2,2)\quad T(0,0,1) = (3,3)$

$(x,y,z) = a_1.\vec{v_1} + a_2\,\vec{v_2}$

$a_1.(0,1,1) + a_2(0,0,1)$

$= (0,\ a_1,\ a_1+a_2)$

$$\begin{cases} a_1 = y \\ a_1+a_2 = z \end{cases} \qquad a_2 = z - \underset{y}{\underset{\sim}{a_1}}$$

$T(x+y) = y.T(\vec{v_1}) + (z-y)\,T(\vec{v_2})$

$y(2,2) + (z-y)(3,3)$

$(2y,\ 2y) + (3z-3y,\ 3z-3y)$

$\boxed{T(x,y,z) = (-y+3z,\ -y+3z)}$

b) $T(1,0,0)$ e

$$\begin{cases} -y+3z = 0 \\ -y+3z = 0 \end{cases} \quad \begin{aligned} x&=0 \\ y&=0 \end{aligned} \quad z = 0 \quad \boxed{\text{Logo}: (0,0)}$$

$T(0,1,0)$

$$\begin{cases} -y+3z = 0 \\ -y+3z = 0 \end{cases} \quad \begin{aligned} -1+3.0&=0 \\ -1+3.0&=0 \end{aligned} \quad \boxed{\text{logo} = (-1,-1)}$$

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9. $T:\mathbb{R}^3\to\mathbb{R}^2\quad T(1,1,1) = (1,2)\quad T(1,1,0) = (2,3)\quad T(1,0,0) = (3,4)$

a) $T(x,y,z)$

$(x,y,z) = a_1.(1,1,1) + a_2(1,1,0) + a_3(1,0,0)$

$(a_1+a_2+a_3,\ a_1+a_2,\ a_1)$

$$\begin{cases} x = a_1+a_2+a_3 \\ y = a_1+a_2 \\ z = a_1 \end{cases} \qquad \begin{aligned} a_2 &= -a_1+y \\ a_2 &= y-z \end{aligned}$$

$T(x+y+z) = z\,T(\vec{v_1}) + (y-z)\,T(\vec{v_2}) + (x-y).T(\vec{v_3})$

$T(x+y+z) = z(1,2) + (y-z).(2,3) + (x-y).(3,4)$

$T(x+y+z) = (z,2z) + (2y-2z,\ 3y-3z) + (3x-3y,\ 4x-4y)$

$\boxed{T(x,y,z) = (-z+3x-y,\ -z-y+4x)}$

b) $v \in \mathbb{R}^3\quad T(v) = (-3,-2)$

$$-\begin{cases} \cancel{-z}+3x\cancel{-y} = -3 \\ \cancel{-z}\cancel{-y}+4x = -2 \end{cases}$$

$0+0-x = -1$

$x = 1$

$-z+3-y = -3$

$-y = -6+z$

$y = 6-z$

$z = z$

$\boxed{v = (1,\ 6-z,\ z)}$

c) $v \in \mathbb{R}^3\quad T(v) = (0,0)$

$$-\begin{cases} \cancel{-z}+3x\cancel{-y} = 0 \\ \cancel{-z}\cancel{-y}+4x = 0 \end{cases}$$

$0+0-x = 0$

$x = 0$

$-z-y = 0$

$y = -z$

$z = z$

$\boxed{v = (0,\ -z,\ z)}$

10. $T: \mathbb{R}^3 \to \mathbb{R}^3$ $T(1,0,0) = (0,2,0)$ $T(0,1,0) = (0,0,-2)$ $T(0,0,1) = (-1,0,3)$ $T(x,y,z)$ $v \in \mathbb{R}^3$ $T(v) = (5,4,-9)$

$(x,y,z) = a_1.(1,0,0) + a_2(0,1,0) + a_3(0,0,1)$

$= a_1, a_2, a_3$

$\begin{cases} a_1 = x \\ a_2 = y \\ a_3 = z \end{cases}$

$x.\vec{v_1} + y.\vec{v_2} + z\vec{v_3}$

$T(x+y+z) = x\,T(\vec{v_1}) + y\,T(\vec{v_2}) + z\,T(\vec{v_3})$

$T(x+y+z) = x.(0,2,0) + y.(0,0,-2) + z(-1,0,3)$

$T(x+y+z) = (0,2x,0) + (0,0,-2y) + (-z,0,3z)$

$\boxed{T(x,y,z) = (-z, 2x, -2y+3z)}$

$T(x) = (5,4,-9)$

$\begin{cases} -z = 5 & z = -5 \\ 2x = 4 & x = 2 \\ -2y + 3z = -9 & y = -3 \end{cases}$

$\boxed{v = (2,-3,-5)}$

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11. $T: P_2 \to P_2$ $T(1) = x$ $T(x) = 1 - x^2$ e $T(x^2) = x + 2x^2$

$T(a+bx+cx^2) = aT(1) + bT(x) + CT(x^2)$

$a(x) + b(1-x^2) + C(x+2x^2)$

$ax + a_2x + b + (-bx^2 + 2cx^2)$

$\boxed{T(a+bx+cx^2) = (a+c)x + b + (-b+2c)x^2}$